IEOR 221 Assignment 9

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1. We used a CRR tree with 10,000 time steps. For BBSR, we applied Black–Scholes smoothing and Richardson extrapolation, using $2V_{20,000} - V_{10,000}$ from trees with 10,000 and 20,000 time steps. Delta, gamma, and theta are computed from the first two levels of each tree, with the same extrapolation applied for BBSR. Vega and rho are computed by central finite differences with absolute bumps of 10^{-4}.

Quantity	CRR	BBSR
NPV	21.1250	21.1253
Delta	0.5217	0.5217
Gamma	0.0060	0.0060
Theta	-1.0716	-1.0714
Vega	67.1553	67.1565
Rho	85.1443	85.1453

2. For the antithentic method, we used the opposite-sign normal samples as a pair. For the control variate method, we implemented two different control functions, which are the ternimal stock price and the price of corresponding geometric Asian call option.

Method	NPV	MC standard error
Plain MC	5.1914	0.0085
Control variate (terminal stock price)	5.1932	0.0057
Control variate (geometric Asian call)	5.1984	0.0003
Antithetic MC	5.1966	0.0069
Antithetic + control variate (terminal stock price)	5.1981	0.0049
Antithetic + control variate (geometric Asian call)	5.1987	0.0003

Each method uses 2^{20} simulated paths, the fixed seed 0, and is computed on CUDA. For antithetic methods, these paths form 2^{19} pairs. The price of the vanilla call option with the same strike and maturity is 8.8249.